

Cyprus International University.

Faculty of Computer Engineering.

CMPE343 Database Management Systems and
Programming-I

Project: Ercan Airport Management System.

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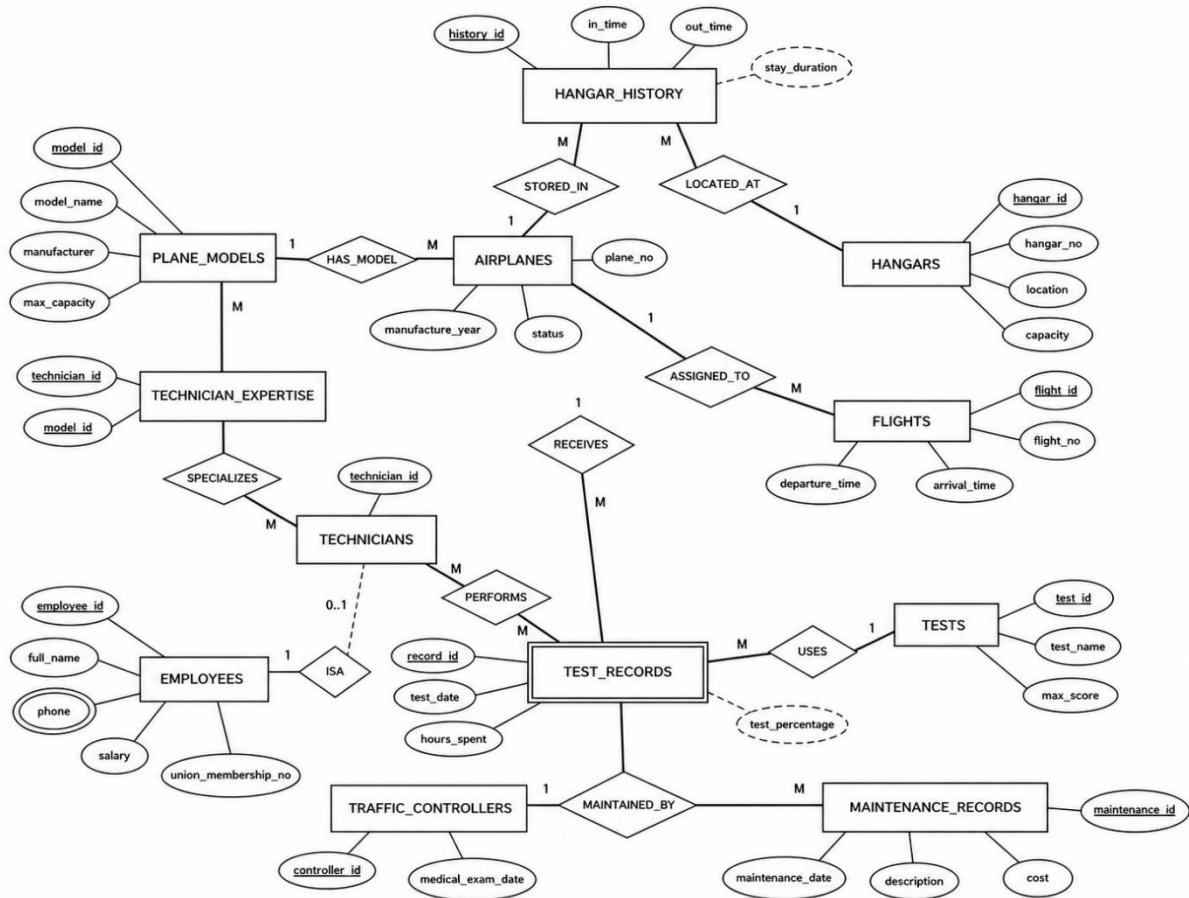
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Introduction

Ercan Airport Management System is a fully MySQL-based database system that stores, manipulates and retrieves data of plane, flight, hangars and employees of its own. The system divides the employees into two parts, the technicians and the traffic controllers. Planes types, number, models' numbers and models' names are also stored in our database. There are some additional features like hangars, hangars history for in-n-out planes monitoring, and we also have maintenance and testing data stored for safety measurements.

ERD



Relational Data Model

PLANE_MODELS(model_id PK,model_name,manufacturer,max_capacity)

AIRPLANES(plane_id PK,plane_no,model_id FK → PLANE_MODELS(model_id),manufacture_year,status)

HANGARS(hangar_id PK,hangar_no,location,capacity)

EMPLOYEES(employee_id PK,full_name,phone,salary,union_membership_no)

TECHNICIANS(technician_id PK,employee_id FK → EMPLOYEES(employee_id))

TRAFFIC_CONTROLLERS(controller_id PK,employee_id FK → EMPLOYEES(employee_id),medical_exam_date)

HANGAR_HISTORY(history_id PK,plane_id FK → AIRPLANES(plane_id),hangar_id FK →
HANGARS(hangar_id),in_time,out_time)

TECHNICIAN_EXPERTISE(technician_id PK, FK → TECHNICIANS(technician_id),model_id PK, FK →
PLANE_MODELS(model_id))

TESTS(test_id PK,test_name, max_score)

TEST_RECORDS(record_id PK,plane_id FK → AIRPLANES(plane_id),technician_id FK →
TECHNICIANS(technician_id),test_id FK → TESTS(test_id),test_date,hours_spent,score)

FLIGHTS(flight_id PK,plane_id FK → AIRPLANES(plane_id),controller_id FK →
TRAFFIC_CONTROLLERS(controller_id),flight_no,departure_airport,arrival_airport,departure_time,arrival_time)

MAINTENANCE_RECORDS(maintenance_id PK,plane_id FK → AIRPLANES(plane_id), technician_id FK →
TECHNICIANS(technician_id),maintenance_date,description,cost)

DDL

```
CREATE TABLE plane_models (  
    model_id INT PRIMARY KEY,  
    model_name VARCHAR(100),  
    manufacturer VARCHAR(100),  
    max_capacity INT  
);
```

```
CREATE TABLE airplanes (  
    plane_id INT PRIMARY KEY,  
    plane_no VARCHAR(20),  
    model_id INT,  
    manufacture_year INT,  
    status VARCHAR(50),  
    FOREIGN KEY (model_id) REFERENCES plane_models(model_id)  
);
```

```
CREATE TABLE hangars (  
    hangar_id INT PRIMARY KEY,  
    hangar_no VARCHAR(20),  
    location VARCHAR(100),  
    capacity INT  
);
```

```
CREATE TABLE employees (  
    employee_id VARCHAR(20) PRIMARY KEY,  
    full_name VARCHAR(100),  
    phone VARCHAR(20),  
    salary DECIMAL(10,2),  
    union_membership_no VARCHAR(50)  
);
```

```
CREATE TABLE hangar_history (  
    history_id INT PRIMARY KEY,  
    plane_id INT,  
    hangar_id INT,  
    in_time DATETIME,  
    out_time DATETIME,  
    FOREIGN KEY (plane_id) REFERENCES airplanes(plane_id),  
    FOREIGN KEY (hangar_id) REFERENCES hangars(hangar_id)  
);
```

```
CREATE TABLE technicians (  
    technician_id INT PRIMARY KEY,  
    employee_id VARCHAR(20),  
    FOREIGN KEY (employee_id) REFERENCES employees(employee_id)  
);
```

```
CREATE TABLE tests (  
    test_id INT PRIMARY KEY,  
    test_name VARCHAR(100),  
    max_score INT  
);
```

```
CREATE TABLE technician_expertise (  
    technician_id INT,  
    model_id INT,  
    PRIMARY KEY (technician_id, model_id),  
    FOREIGN KEY (technician_id) REFERENCES technicians(technician_id),  
    FOREIGN KEY (model_id) REFERENCES plane_models(model_id)  
);
```

```

CREATE TABLE traffic_controllers (
    controller_id INT PRIMARY KEY,
    employee_id VARCHAR(20),
    medical_exam_date DATE,
    FOREIGN KEY (employee_id) REFERENCES employees(employee_id)
);

CREATE TABLE flights (
    flight_id INT PRIMARY KEY,
    plane_id INT,
    controller_id INT,
    flight_no VARCHAR(20),
    departure_airport VARCHAR(100),
    arrival_airport VARCHAR(100),
    departure_time DATETIME,
    arrival_time DATETIME,
    FOREIGN KEY (plane_id) REFERENCES airplanes(plane_id),
    FOREIGN KEY (controller_id) REFERENCES traffic_controllers(controller_id)
);

CREATE TABLE maintenance_records (
    maintenance_id INT PRIMARY KEY,
    plane_id INT,
    technician_id INT,
    maintenance_date DATE,
    description TEXT,
    cost DECIMAL(10,2),
    FOREIGN KEY (plane_id) REFERENCES airplanes(plane_id),
    FOREIGN KEY (technician_id) REFERENCES technicians(technician_id)
);

```

```
CREATE TABLE test_records (  
    record_id INT PRIMARY KEY,  
    plane_id INT,  
    technician_id INT,  
    test_id INT,  
    test_date DATE,  
    hours_spent DECIMAL(5,2),  
    score INT,  
    FOREIGN KEY (plane_id) REFERENCES airplanes(plane_id),  
    FOREIGN KEY (technician_id) REFERENCES technicians(technician_id),  
    FOREIGN KEY (test_id) REFERENCES tests(test_id)  
);
```

DML

```
INSERT INTO plane_models VALUES  
  
(1, 'Boeing 737', 'Boeing', 189),  
(2, 'Airbus A320', 'Airbus', 180),  
(3, 'Boeing 777', 'Boeing', 396),  
(4, 'Airbus A330', 'Airbus', 300),  
(5, 'Embraer E190', 'Embraer', 114),  
(6, 'Cessna 172', 'Cessna', 4),  
(7, 'Boeing 787', 'Boeing', 335),  
(8, 'Airbus A350', 'Airbus', 350),  
(9, 'Bombardier CRJ900', 'Bombardier', 90),  
(10, 'ATR 72', 'ATR', 72);
```


INSERT INTO airplanes VALUES

(1, 'ERC101', 1, 2018, 'Active'),
(2, 'ERC102', 2, 2020, 'Maintenance'),
(3, 'ERC103', 3, 2016, 'Active'),
(4, 'ERC104', 4, 2019, 'Active'),
(5, 'ERC105', 5, 2021, 'Inspection'),
(6, 'ERC106', 6, 2015, 'Active'),
(7, 'ERC107', 7, 2022, 'Active'),
(8, 'ERC108', 8, 2023, 'Active'),
(9, 'ERC109', 9, 2017, 'Maintenance'),
(10, 'ERC110', 10, 2014, 'Active');

INSERT INTO hangars VALUES

(1, 'H1', 'North Zone', 5),
(2, 'H2', 'South Zone', 8),
(3, 'H3', 'East Wing', 6),
(4, 'H4', 'West Wing', 4),
(5, 'H5', 'Cargo Area', 10),
(6, 'H6', 'Repair Center', 7),
(7, 'H7', 'VIP Area', 2),
(8, 'H8', 'Maintenance Zone', 9),
(9, 'H9', 'Overflow Area', 12),
(10, 'H10', 'Remote Hangar', 3);

INSERT INTO employees VALUES

('EID101', 'Ahmed Hassan', '0551234567', 3500, 'UN1001'),
('EID102', 'Ali Salem', '0552223344', 4200, 'UN1002'),
('EID103', 'Murat Yildiz', '0553334455', 5000, 'UN1003'),
('EID104', 'Mohammed Yasin', '0554445566', 3900, 'UN1004'),
('EID105', 'Khaled Nasser', '0555556677', 4600, 'UN1005'),
('EID106', 'Arda Demir', '0556667788', 5200, 'UN1006'),
('EID107', 'Tariq Saleh', '0557778899', 4100, 'UN1007'),
('EID108', 'Yousef Ibrahim', '0558889900', 3700, 'UN1008'),
('EID109', 'Omar Khalil', '0559991122', 4800, 'UN1009'),
('EID110', 'Hakan Kaya', '0551112233', 5100, 'UN1010'),
('EID111', 'Ibrahim Mansour', '0552224455', 4700, 'UN1011');

INSERT INTO technicians VALUES

(1, 'EID101'),
(2, 'EID102'),
(3, 'EID104'),
(4, 'EID105'),
(5, 'EID106'),
(6, 'EID107'),
(7, 'EID108'),
(8, 'EID103');

INSERT INTO tests VALUES

(1, 'Engine Test', 100),
(2, 'Hydraulic Test', 100),
(3, 'Electrical Systems', 100),
(4, 'Landing Gear Check', 100),
(5, 'Fuel System Test', 100),
(6, 'Cabin Pressure Test', 100),
(7, 'Navigation Test', 100),
(8, 'Emergency Equipment Check', 100),
(9, 'Wing Structural Test', 100),
(10, 'Avionics Test', 100);

INSERT INTO hangar_history VALUES

(1, 1, 1, '2026-05-01 08:00:00', '2026-05-02 10:00:00'),
(2, 2, 2, '2026-05-02 09:00:00', NULL),
(3, 3, 3, '2026-05-03 10:00:00', '2026-05-04 12:00:00'),
(4, 4, 4, '2026-05-04 11:00:00', NULL),
(5, 5, 5, '2026-05-05 07:00:00', '2026-05-06 09:00:00'),
(6, 6, 6, '2026-05-06 08:30:00', NULL),
(7, 7, 7, '2026-05-07 09:15:00', '2026-05-08 11:45:00'),
(8, 8, 8, '2026-05-08 10:00:00', NULL);

INSERT INTO traffic_controllers VALUES

(1, 'EID109', '2026-03-15'),
(2, 'EID110', '2026-04-10'),
(3, 'EID111', '2026-02-20');

INSERT INTO maintenance_records VALUES

(1, 1, 1, '2026-05-01', 'Engine maintenance', 500.00),
(2, 2, 2, '2026-05-02', 'Hydraulic system check', 300.00),
(3, 3, 8, '2026-05-03', 'Electrical systems inspection', 400.00),
(4, 4, 3, '2026-05-04', 'Landing gear maintenance', 600.00),
(5, 5, 4, '2026-05-05', 'Fuel system maintenance', 700.00),
(6, 6, 5, '2026-05-06', 'Cabin pressure maintenance', 250.00),
(7, 7, 6, '2026-05-07', 'Navigation system repair', 450.00),
(8, 8, 7, '2026-05-08', 'Emergency equipment service', 350.00),
(9, 9, 1, '2026-05-09', 'Wing structure maintenance', 800.00),
(10, 10, 2, '2026-05-10', 'Avionics repair', 650.00);

INSERT INTO technician_expertise VALUES

(1, 1),
(1, 3),
(2, 2),
(2, 4),
(3, 5),
(3, 10),
(4, 6),
(5, 7),
(5, 8),
(6, 9),
(7, 1),
(7, 2),
(8, 3),
(8, 7);

```
INSERT INTO test_records VALUES
```

```
(1, 1, 1, 1, '2026-05-01', 2.5, 88),  
(2, 2, 2, 2, '2026-05-02', 3.0, 76),  
(3, 3, 8, 3, '2026-05-03', 2.0, 91),  
(4, 4, 3, 4, '2026-05-04', 4.0, 84),  
(5, 5, 4, 5, '2026-05-05', 1.5, 79),  
(6, 6, 5, 6, '2026-05-06', 2.0, 95),  
(7, 7, 6, 7, '2026-05-07', 3.5, 89),  
(8, 8, 7, 8, '2026-05-08', 2.5, 92),  
(9, 9, 1, 9, '2026-05-09', 4.5, 73),  
(10, 10, 2, 10, '2026-05-10', 3.0, 86);
```

```
INSERT INTO flights VALUES
```

```
(1, 1, 1, 'ER1001', 'Ercan Airport', 'Istanbul Airport', '2026-05-01 09:00:00', '2026-05-01 10:30:00'),  
(2, 2, 2, 'ER1002', 'Ercan Airport', 'Ankara Esenboga', '2026-05-02 11:00:00', '2026-05-02 12:20:00'),  
(3, 3, 3, 'ER1003', 'Ercan Airport', 'London Heathrow', '2026-05-03 08:30:00', '2026-05-03 13:30:00'),  
(4, 4, 1, 'ER1004', 'Ercan Airport', 'Dubai International', '2026-05-04 14:00:00', '2026-05-04 18:00:00'),  
(5, 5, 2, 'ER1005', 'Ercan Airport', 'Antalya Airport', '2026-05-05 16:00:00', '2026-05-05 17:10:00'),  
(6, 6, 3, 'ER1006', 'Ercan Airport', 'Izmir Adnan Menderes', '2026-05-06 07:45:00', '2026-05-06 09:00:00'),  
(7, 7, 1, 'ER1007', 'Ercan Airport', 'Frankfurt Airport', '2026-05-07 10:15:00', '2026-05-07 14:00:00'),  
(8, 8, 2, 'ER1008', 'Ercan Airport', 'Paris Charles de Gaulle', '2026-05-08 13:20:00', '2026-05-08 17:10:00'),  
(9, 9, 3, 'ER1009', 'Ercan Airport', 'Beirut Airport', '2026-05-09 12:00:00', '2026-05-09 13:00:00'),  
(10, 10, 1, 'ER1010', 'Ercan Airport', 'Cairo Airport', '2026-05-10 15:30:00', '2026-05-10 17:20:00');
```

Queries

--1)Display Plane Number and Model Name of all airplanes in the database.

```
SELECT a.plane_no , pm.model_name
```

```
FROM airplanes a
```

```
JOIN plane_models pm ON a.model_id = pm.model_id;
```

plane_no	model_name
ERC101	Boeing 737
ERC102	Airbus A320
ERC103	Boeing 777
ERC104	Airbus A330
ERC105	Embraer E190
ERC106	Cessna 172
ERC107	Boeing 787
ERC108	Airbus A350
ERC109	Bombardier CRJ900
ERC110	ATR 72

--2)List the full names and phone numbers of all employees who are technicians.

```
SELECT e.full_name, e.phone
```

```
FROM employees e
```

```
JOIN technicians t ON e.employee_id = t.employee_id;
```

full_name	phone
Ahmed Hassan	0551234567
Ali Salem	0552223344
Murat Yildiz	0553334455
Mohammed Yasin	0554445566
Khaled Nasser	0555556677
Arda Demir	0556667788
Tariq Saleh	0557778899
Yousef Ibrahim	0558889900

--3)List the full name and phone number of all employees who are traffic controllers.

```
SELECT e.full_name, e.phone
```

```
FROM employees e
```

```
JOIN traffic_controllers tc ON e.employee_id = tc.employee_id;
```

full_name	phone
Omar Khalil	0559991122
Hakan Kaya	0551112233
Ibrahim Mansour	0552224455

--4)List Model Name and Hnager Number of all planes that are currently in a hangar.

```
SELECT pm.model_name, h.hangar_no
```

```
FROM airplanes a
```

```
JOIN plane_models pm ON a.model_id = pm.model_id
```

```
JOIN hangar_history hh ON a.plane_id = hh.plane_id
```

```
JOIN hangars h ON hh.hangar_id = h.hangar_id
```

```
WHERE hh.out_time IS NULL;
```

model_name	hangar_no
Airbus A320	H2
Airbus A330	H4
Cessna 172	H6
Airbus A350	H8

--5)List the full name of all technicians who have expertise in the plane model with the highest maximum capacity.

```
SELECT e.full_name
```

```
FROM employees e
```

```
JOIN technicians t ON e.employee_id = t.employee_id
```

```
JOIN technician_expertise te ON t.technician_id = te.technician_id
```

```
JOIN plane_models pm ON te.model_id = pm.model_id
```

```
WHERE pm.max_capacity = (SELECT MAX(max_capacity) FROM plane_models);
```

full_name
Ahmed Hassan
Murat Yildiz

--6)Display every record of flights in the database.

```
SELECT *
```

```
FROM flights;
```

flight_id	plane_id	controller_id	flight_no	departure_airport	arrival_airport	departure_time	arrival_time
1	1	1	ER1001	Ercan Airport	Istanbul Airport	2026-05-01 09:00:00	2026-05-01 10:30:00
2	2	2	ER1002	Ercan Airport	Ankara Esenboga	2026-05-02 11:00:00	2026-05-02 12:20:00
3	3	3	ER1003	Ercan Airport	London Heathrow	2026-05-03 08:30:00	2026-05-03 13:30:00
4	4	1	ER1004	Ercan Airport	Dubai International	2026-05-04 14:00:00	2026-05-04 18:00:00
5	5	2	ER1005	Ercan Airport	Antalya Airport	2026-05-05 16:00:00	2026-05-05 17:10:00
6	6	3	ER1006	Ercan Airport	Izmir Adnan Menderes	2026-05-06 07:45:00	2026-05-06 09:00:00
7	7	1	ER1007	Ercan Airport	Frankfurt Airport	2026-05-07 10:15:00	2026-05-07 14:00:00
8	8	2	ER1008	Ercan Airport	Paris Charles de Gaulle	2026-05-08 13:20:00	2026-05-08 17:10:00
9	9	3	ER1009	Ercan Airport	Beirut Airport	2026-05-09 12:00:00	2026-05-09 13:00:00
10	10	1	ER1010	Ercan Airport	Cairo Airport	2026-05-10 15:30:00	2026-05-10 17:20:00

--7) List the Model name of planes with a score of 85 or higher in any test.

```
SELECT DISTINCT d.model_name
FROM plane_models d
JOIN airplanes a ON d.model_id = a.model_id
JOIN test_records tr ON a.plane_id = tr.plane_id
WHERE tr.score >= 85;
```

model_name
Boeing 737
Boeing 777
Cessna 172
Boeing 787
Airbus A350
ATR 72

--8) List Employees with salary higher than 5000.

```
SELECT full_name, salary
FROM employees
WHERE salary > 5000;
```

full_name	salary
Arda Demir	5200.00
Hakan Kaya	5100.00

--9) List Full names of traffic controllers who have had a medical exam in the last 6 months.

```
SELECT e.full_name
FROM employees e
```

JOIN traffic_controllers tc ON e.employee_id = tc.employee_id

WHERE tc.medical_exam_date >= DATE_SUB(CURDATE(), INTERVAL 6 MONTH);

```
+-----+
| full_name |
+-----+
| Omar Khalil |
| Hakan Kaya |
| Ibrahim Mansour |
+-----+
```

--10) List the full names of technicians who have expertise in the plane model with the lowest maximum capacity.

SELECT e.full_name

FROM employees e

JOIN technicians t ON e.employee_id = t.employee_id

JOIN technician_expertise te ON t.technician_id = te.technician_id

JOIN plane_models pm ON te.model_id = pm.model_id

WHERE pm.max_capacity = (SELECT MIN(max_capacity) FROM plane_models);

```
+-----+
| full_name |
+-----+
| Khaled Nasser |
+-----+
```

--11) Display the test that took more than 3 hours and had a score less than 80.

SELECT t.test_name, tr.hours_spent, tr.score

FROM tests t

JOIN test_records tr ON t.test_id = tr.test_id

WHERE tr.hours_spent > 3 AND tr.score < 80;

test_name	hours_spent	score
Wing Structural Test	4.50	73

--12) List the flights that took more than 5 hours.

```
SELECT flight_no, departure_time, arrival_time,
       TIMESTAMPDIFF(HOUR, departure_time, arrival_time) AS flight_time_hours
FROM flights
WHERE TIMESTAMPDIFF(HOUR, departure_time, arrival_time) > 3;
```

flight_no	departure_time	arrival_time	flight_time_hours
ER1003	2026-05-03 08:30:00	2026-05-03 13:30:00	5
ER1004	2026-05-04 14:00:00	2026-05-04 18:00:00	4

--13) Display the hangar with the biggest capacity.

```
SELECT hangar_no, capacity, location
FROM hangars
WHERE capacity = (SELECT MAX(capacity) FROM hangars);
```

hangar_no	capacity	location
H9	12	Overflow Area

--14) Display the full name of the traffic controller who controlled the flight with the longest duration.

```
SELECT e.full_name, f.flight_no, TIMESTAMPDIFF(HOUR, f.departure_time,
f.arrival_time) AS flight_duration_hours
FROM employees e
JOIN traffic_controllers tc ON e.employee_id = tc.employee_id
```

JOIN flights f ON tc.controller_id = f.controller_id

ORDER BY flight_duration_hours DESC

LIMIT 1;

full_name	flight_no	flight_duration_hours
Ibrahim Mansour	ER1003	5

--15) Display the full name of the technician who performed the most maintenance records.

SELECT e.full_name, COUNT(m.maintenance_id) AS maintenance_count

FROM employees e

JOIN technicians t ON e.employee_id = t.employee_id

JOIN maintenance_records m ON t.technician_id = m.technician_id

GROUP BY e.full_name

ORDER BY maintenance_count DESC

LIMIT 1;

full_name	maintenance_count
Ahmed Hassan	2